

In T.849/11 the board summarised case law on the requirements of Art. 84 EPC with regard to the characterisation of an invention by parameters. The board stated that: i. the claims must be clear in themselves when read by the skilled person; and ii. the method for measuring a parameter (or at least a reference thereto) must appear completely in the claim itself (see T.1156/01, T.412/02, T.908/04, T.555/05, T.1497/08), and that iii. the applicant who chooses to define the scope of the claim by parameters should ensure that a skilled person can easily and unambiguously verify whether he is working inside or outside the scope of the claim. The board further stated that the requirements of Art. 84 EPC can still be met if it can be convincingly shown that: i. the method to be employed belongs to the skilled person's common general knowledge, or ii. all the methodologies known in the relevant technical field for determining this parameter yield the same result within the appropriate limit of measurement accuracy (see T.1156/01).

In T.29/05 it was found that although different experimental protocols might be applied in the case in point for assessing hybridisation under stringent conditions, they were usual in the art (following T.1084/00).

In T.307/06 the claims contained the added feature "and a T_g of less than 25°C". The board noted that there were different methods for determining T_g (glass transition temperature). However, the mere fact that several methods existed did not render the claims unclear if at least one of the following conditions was met: (a) the different methods yield essentially the same T_g values for the same material, or (b) the person skilled in the art had associated the range of T_g values mentioned in claim 1 at issue with only one standard method of measurement. Neither condition (a) nor (b) was fulfilled. The board concluded that this left doubt as to which subject-matter was covered by claim 1, and thus rendered said claim unclear (see T.728/98, OJ 2001, 319; T.306/13).

In T.2676/16 the board disagreed with the examining division that the parameter "an outcome of a pseudo-random hopping" of the method of claim 1 was not clear because it could have many interpretations to a skilled person in the field of telecommunications. In context, the term "outcome" had the same meaning as "result". It was clear that by means of the "pseudo-random hopping" operation a scalar number was to be produced. The meaning of "pseudo-random" was also clear for the skilled reader. The fact that a feature could be implemented in various ways did not necessarily render this feature unclear.

In a number of decisions the boards stress that it must be clear to the skilled reader from the claim itself how the parameters are to be determined, unless it can be demonstrated that the skilled reader would immediately know which method and conditions to apply even without any such indication in the claim.

In T.412/02 the board held that the unambiguous characterisation in a claim of a product by parameters or by a mathematical relation between parameters necessarily required that each parameter could be clearly and reliably determined. In the board's view, it followed that the knowledge of the method and conditions of determination of the parameter was necessary for the unambiguous definition of the parameters and, as a consequence, for the unambiguous definition of a mathematical relation between them. Thus, in order to allow the matter for which protection was sought to be defined, it had to

be clear from the claim itself when being read by the person skilled in the art exactly how the parameters should be determined. This would, as a rule, imply that the method of determination and the conditions of measurement which might have an influence on the value of the parameter should be indicated in the claim, either expressly or, if appropriate, by way of reference to the description according to R. 29(6) EPC 1973. Such indication would only become superfluous provided it could be shown that the skilled person would know from the outset which method and conditions to employ. See also T 1156/01.

In T 1819/07 the board held that the parameter "average particle size" rendered the claim unclear as neither the type of average (volume, surface, number) nor a method for determining it was indicated in the claims. See also T 967/08 and T 45/10.

In T 992/02 the board took the view that whilst noting the general principle that the claims must be clear per se, the non-inclusion in the claim of the method whereby one of the parameters characterising the composition claimed (level of formation of volatiles) could be measured was justified by the requirement that the claims be concise. It considered that such a conclusion could be reached in the case in hand because the method was clearly identified in the description and did not give rise to any ambiguity.

In T 2086/11 the parameter "mean aspect ratio" represented the mean of the individual aspect ratios, defined as the ratio of the major axis to thickness, of 50 aluminium flake particles having major axes of not more than 10 µm. The board held that this parameter did not enable to distinguish which subject-matter is covered by the claims and which is not. According to the method of measuring explained in the description, the investigated sample contained thousands of particles having a major axis of not more than 10 µm and the method described in the application did not give any precise indication of how to select such 50 particles. One operator, by selecting at random one group of 50 particles, could measure a mean aspect ratio within the claim whilst another operator, by selecting a different group of 50 particles, could find a value of mean aspect ratio outside the claim. The parameter "mean aspect ratio" was thus by itself insufficient to characterise the claimed product in a reliable way and did not comply with Art. 84 EPC 1973.

3.6. Unspecified features and relative qualities

Unspecified features are permitted as follows:

- (a) To avoid an undue limitation of the scope of the claim where the person skilled in the art can verify the result by tests involving nothing more than trial and error (T 88/87).
- (b) To render the meaning of the feature intelligible from the actual claim.

Thus, the objection in T 487/89 was based on the fact that the claim had indicated both tenacity and toughness with a lower but no upper limit. The opposition division took the view that such "open-ended" parameters were always objectionable if they related to an inherently desirable characteristic. The board, however, held that whether the absence of an upper or lower limit was acceptable in a claim in any individual case depended on all the surrounding circumstances. Where, as in the case in question, the claim sought to

embrace values as high as could be attained above a specified minimum level, given the other parameters of the claim, then such open-ended parameters could not normally be objected to. This decision was confirmed by T 129/88 (OJ 1993, 598), T 87/84, T 92/84, T 136/84, T 297/90 and T 1018/05, although in the last two cases a role was played by the question of sufficient disclosure (see also T 989/95, in which T 487/89 was agreed with but not applied). In T 586/97, on the other hand, the board found that the absence from any independent claim of any upper limit on the amount of a particular detrimental but necessary component in the chemical composition claimed was at variance with the aim of the invention as set out in the description, namely, to decrease the percentage of the undesirable ingredient in the claimed composition. The claim was thus so broad it went beyond the scope of the invention as disclosed in the description. Therefore the requirement of Art. 84, second sentence, EPC 1973 was not met.

In T 227/91 (OJ 1994, 491) the claims comprised a coating thickness defined by a formula with two parameters (a) and (t). Parameter (a) represented the thermal diffusivity of the coating means and was therefore a feature inherent in the instrument. Parameter (t) represented the effective pulse time of the laser and was related to the laser operating conditions, not to the structure of either the laser or the instrument. The thickness defined in the claims was still connected with the mode of operation of the laser, that is, with a human factor irrelevant to the instrument per se. The extent of the protection conferred by the subject-matter of the claims was therefore regarded by the board as ambiguous and indefinite.

In T 455/92 independent claim 2 was worded as follows: "Covering [...] for a pressing [...], characterised by: (a) a length which corresponds at least to the circumference of the shell face of the pressing to be covered, (b) an overall breadth sufficient to cover the shell face and both end faces of the pressing, and (c) a fold such that [...] the breadth [...] corresponds to the breadth of the shell face of the pressing [...]". The board did not see this as a breach of Art. 84 EPC 1973, since reference had been made to a physical entity which was known; hence the range of sizes was definable in so far as it excluded packaging materials for other physical entities. Furthermore, it was self-evident to a skilled person in this field that the size of the packaging material (the covering) was dependent on the physical entity to be packed. In T 1020/13, however, the definition of a claimed entity by reference to a second entity that was not part of the claim was deemed a breach of Art. 84 EPC.

In T 860/93 (OJ 1995, 47) the board decided as follows: Where a quality is expressed in a claim as being within a given numerical range, the method for measuring that quality must either be general technical knowledge, so that no explicit description is needed, or a method of measuring that quality needs to be identified (decision T 124/85 followed). In contrast, where a claim specifies a relative quality, in this case that the products should be "water-soluble", it is not normally necessary to identify any method for its determination (for the term "soluble" see also T 785/92, T 939/98, T 125/15). In T 860/95 ("for a long period of time"), T 649/97 ("transparent"), T 1041/98 ("thin plate"), T 193/01 ("thin film"), T 545/01 ("flat"), T 378/02 ("smooth"), T 2367/16 ("overnight"), and T 2402/17 (relationship between the terms "higher" and "lower oscillation susceptibility" and "smaller values"), the boards confirmed that the use of a relative term in a claim may be accepted where the skilled person is able to understand the meaning of this term in a given context. However,

in T. 728/98 ("substantially pure"), T. 174/02 ("[in] quick [succession]"), T. 1640/11 ("value"), T. 842/14 ("accurate"), T. 1963/17 ("substantially simultaneously") and T. 362/17 ("brief [history of spoken words]"), the boards ruled that the relative terms at issue were unclear.

In the application at issue in T. 1888/12, claim 1 referred to undated ISO standards. The board pointed out that such standards were typically revised over time, and the possibility of substantial changes could not be ruled out (see T. 783/05). The appellant's argument that the date to be taken into account was the application's priority date did not convince the board because there was no such indication in the description. In the board's view, the definition of technical features by general reference to ISO standards was not sufficiently clear. However, in T. 2187/09, the board decided that the UNISIG standard mentioned in the claim merely provided context for the invention. Details of the standard were not needed to define the scope of the invention and to enable the skilled person to know if he was working within or outside the claimed scope.

According to T. 3003/18, the clarity of a claim specifying a norm or standard depended on the case, and in particular the claimed subject-matter. The MTRJ connector specified in claim 1 designated a well-known standardised family of connectors having, on the one hand, specific features that could differ among the connectors of the family and could change over time according to different specific versions of the corresponding standard or norm and, on the other hand, a series of common features that ensured a predetermined degree of compatibility and interchangeability between the different connectors of the same family. These connectors were commonly and generally designated "MTRJ connectors" in textbooks, reference books and the like, without reference to any particular version of the corresponding standard or norm. The features of the MTRJ connector relevant in the technical context of claim 1 were the common features, and so the skilled person would understand what was meant in the claimed context by the optic connector being an "MTRJ connector".

3.7. Principles in connection with categories of claim

In T. 688/91 the board found that the fact that claims of different categories were deemed to be independent claims did not preclude reference to other independent claims. In the board's view, however, any such reference had to make perfectly clear whether or not the reference was intended to replace features. In the former case, it had for reasons of clarity to replace all the features, and in the latter case none.

In case T. 410/96 concerning a data processing system the question arose whether it was permissible for a claim to a system to refer back to a claim to a method. The board held that, although R. 29(4) EPC 1973 mentioned a reference only in the context of a claim including all the features of the claim it referred to, this did not imply a contrario that a reference which did not fulfil this condition was necessarily impermissible. The board found that there was no a priori objection under Art. 84 EPC 1973 to an independent claim which partially drew its features from preceding claims of a different category. The essential features were thus incorporated in the disputed claim by reference, the claim thus complying with R. 29(3) EPC 1973 (R. 43(3) EPC).

In T. 1722/11 a claim related to a computer program resident on computer-readable media was defined by reference to a method claim. The execution of the method required co-ordinated interaction between various resources in a communication network. As these resources were at different, mutually remote locations, the board considered it unclear how the method referred to could be executed solely by a (single) computer program when run on a computer. The claim thus lacked clarity.

In decisions T. 418/92 and T. 453/90 the boards confirmed that it was possible for claims to relate to both physical activities and physical entities, in particular where the inventions concerned were fully disclosed as processes to the skilled person only if apparatus features were also included in the definition. In T. 1046/05 too the board confirmed the clarity of an apparatus feature in a process claim.

In T. 426/89 (OJ 1992, 172) the board decided that a claim relating to a "method for operating" a device was not "clear" (Art. 84 EPC 1973) if the claimed features only described how the device worked.

In T. 841/95 the board found that where a claim for an apparatus sought to define the invention by reference to features of the use to which the apparatus was to be put, a lack of clarity could result. However, if the subject-matter of the application was an apparatus and it was not viable to define the apparatus other than by reference to the said use, an apparatus claim directed to a combination of apparatus and use and defining the apparatus unambiguously could be clear.

Including in a device claim "for carrying out a process" a reference back to an earlier claim did not alter the fact that it was an independent (non-process) claim; rather, it limited the claimed device to being suitable to carry out the process referred to (T. 1017/98).

A claim in T. 952/99 contained two different aspects of a use claim: on the one hand it defined the use of a particular physical entity to achieve an 'effect', and on the other it defined its use to produce a product. Although a use claim normally falls into either one category or the other, the board found that such a clear distinction was not possible in this case. The claim contained both application and process features and there was a technical affinity between the additional process step and the 'core' of the invention. Both the application and process features were thus technical features of the invention, so that their combination did not contravene R. 29(1) EPC 1973, the relevant criterion for assessing the extent to which the combination of a 'use' claim and technical process features was allowable from the point of view of clarity. The board drew a distinction with T. 796/91, in which it had been held that the heterogeneous combination of process features and application features rendered the claim fundamentally unclear. That case had concerned the combination of a process claim and a use claim with no process claim at all. In contrast to the case before the board, the process features had apparently had no affinity to the application features, which had made it impossible to subsume the two different types of technical feature under one claim.

In T. 921/11 the board considered that the claims as a whole, construed in the light of the description, were so unclear as to the meaning of the term "system" that the skilled reader